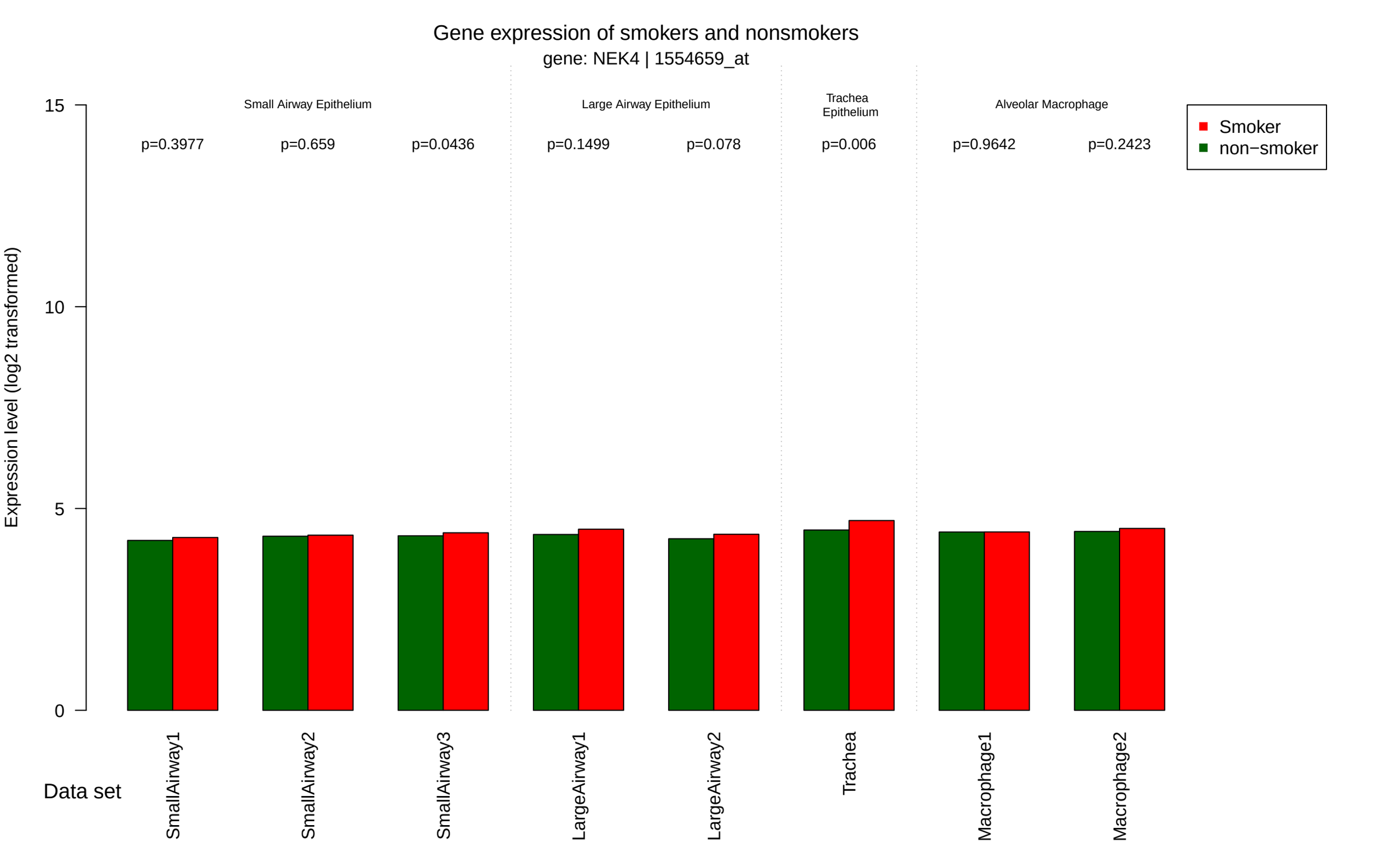
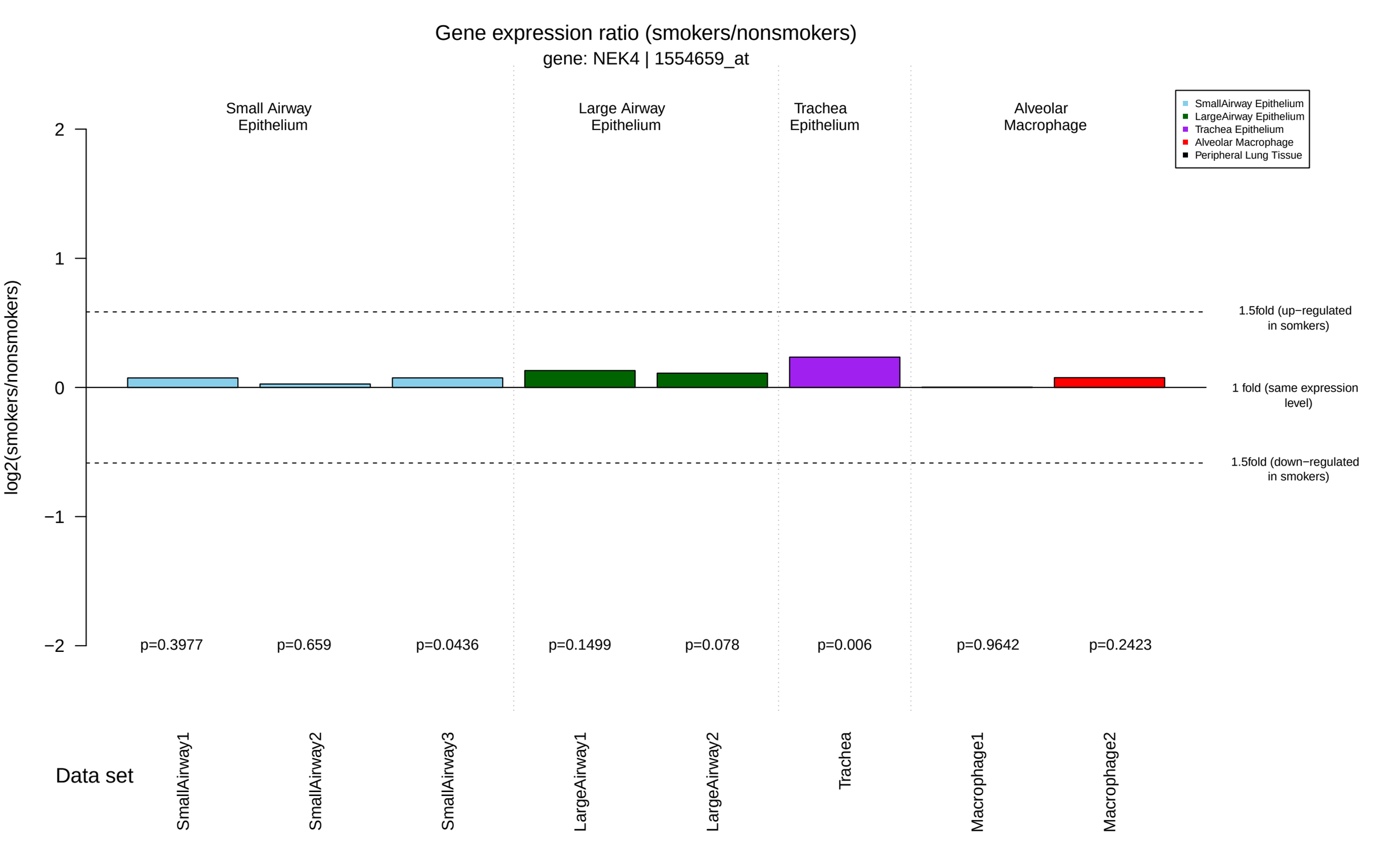
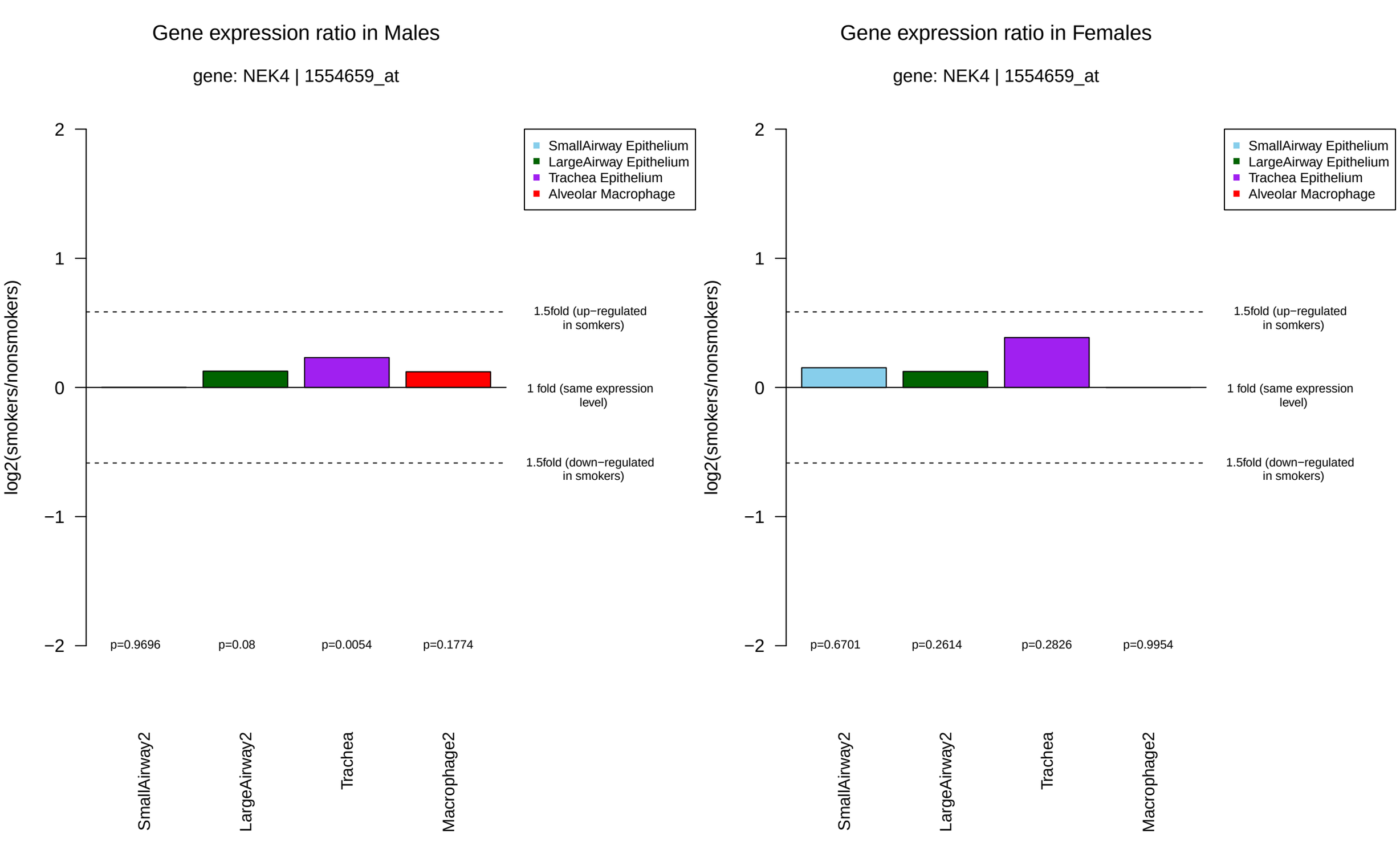
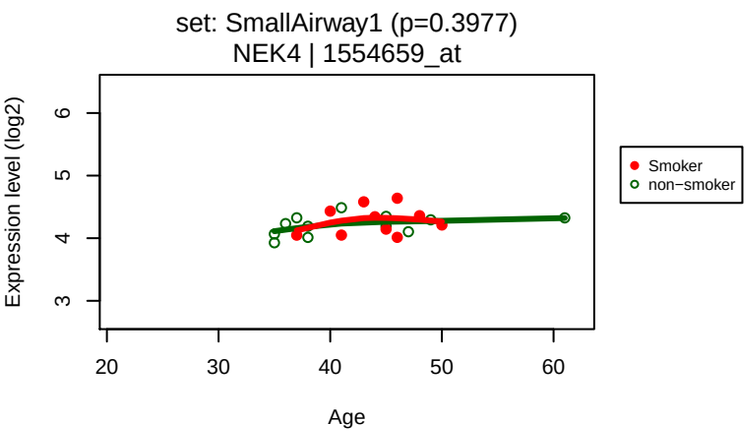
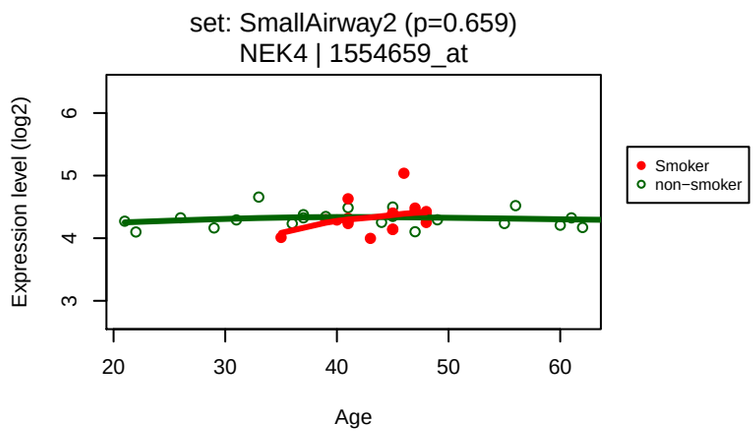
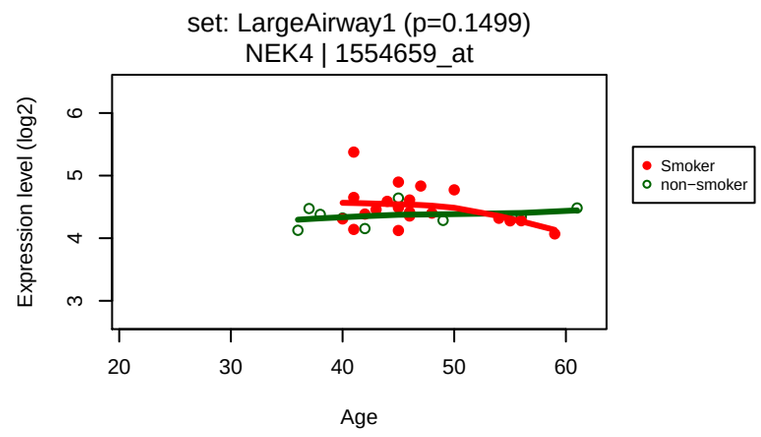

Gene expression of smokers and nonsmokers
gene: NEK4 | 1554659_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3977
p=0.659
p=0.0436
p=0.1499
p=0.078
p=0.006
p=0.9642
p=0.2423
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

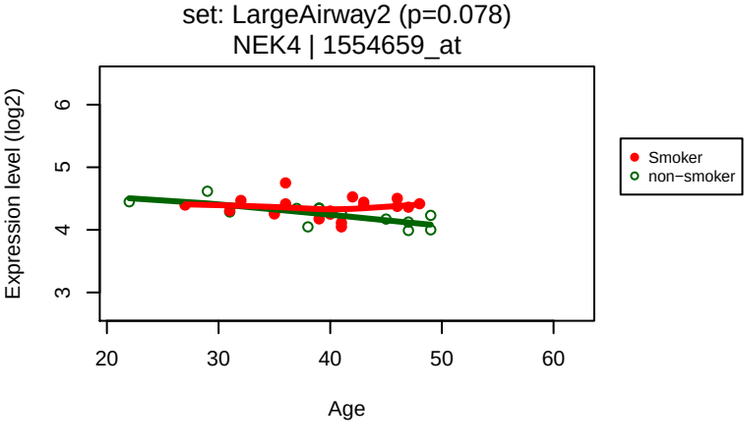

Gene expression ratio (smokers/nonsmokers)
gene: NEK4 | 1554659_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3977
p=0.659
p=0.0436
p=0.1499
p=0.078
p=0.006
p=0.9642
p=0.2423
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

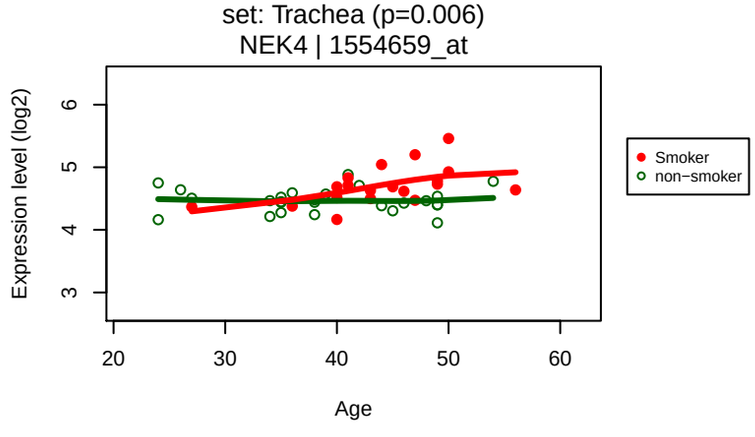

Gene expression ratio in Males
gene: NEK4 | 1554659_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9696
p=0.08
p=0.0054
p=0.1774
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NEK4 | 1554659_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6701
p=0.2614
p=0.2826
p=0.9954
SmallAirway2
LargeAirway2
Trachea
Macrophage2

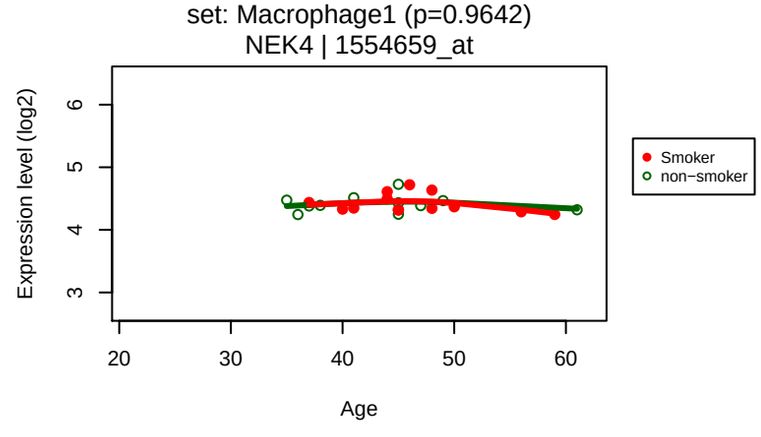

set: SmallAirway1 (p=0.3977)
NEK4 | 1554659_at
Expression level (log2)
Age
Smoker
non−smoker

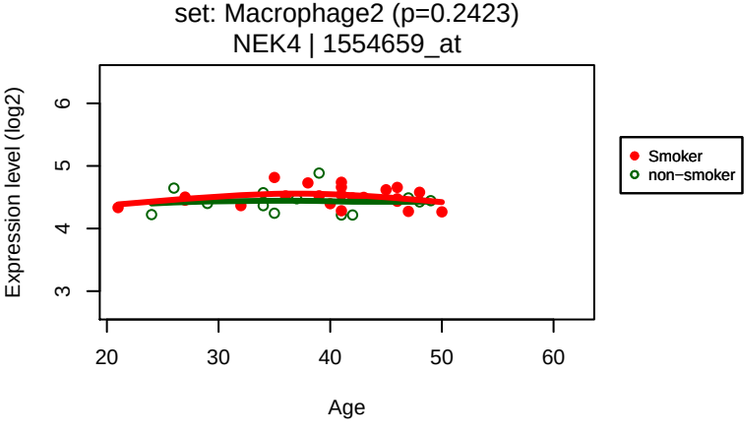

set: SmallAirway2 (p=0.659)
NEK4 | 1554659_at
Expression level (log2)
Age
Smoker
non−smoker

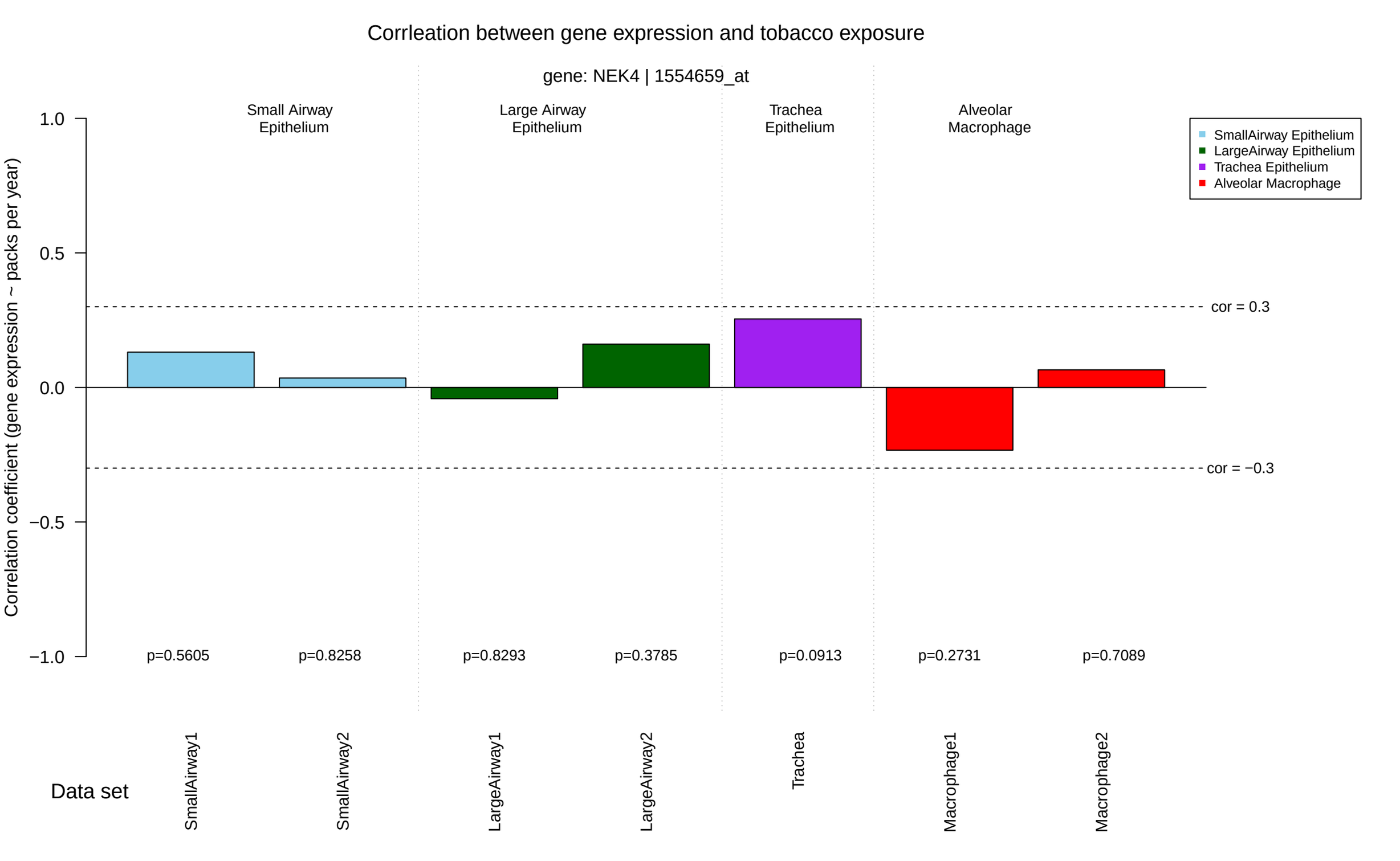

set: LargeAirway1 (p=0.1499)
NEK4 | 1554659_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.078)
NEK4 | 1554659_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.006)
NEK4 | 1554659_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9642)
NEK4 | 1554659_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2423)
NEK4 | 1554659_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NEK4 | 1554659_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5605
p=0.8258
p=0.8293
p=0.3785
p=0.0913
p=0.2731
p=0.7089
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2